



LUMS



Machine Learning



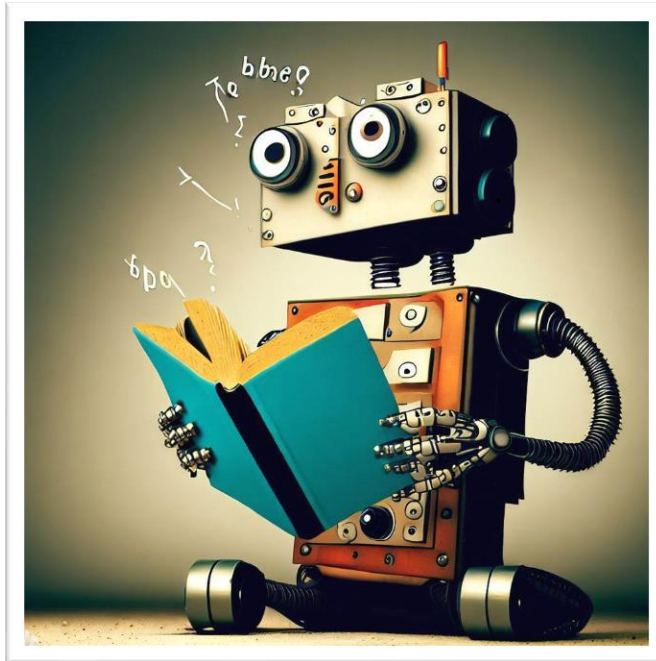
Agha Ali Raza



The Wonderful World of Artificial Intelligence

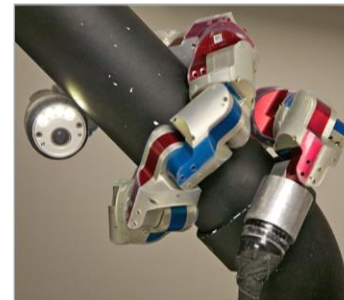


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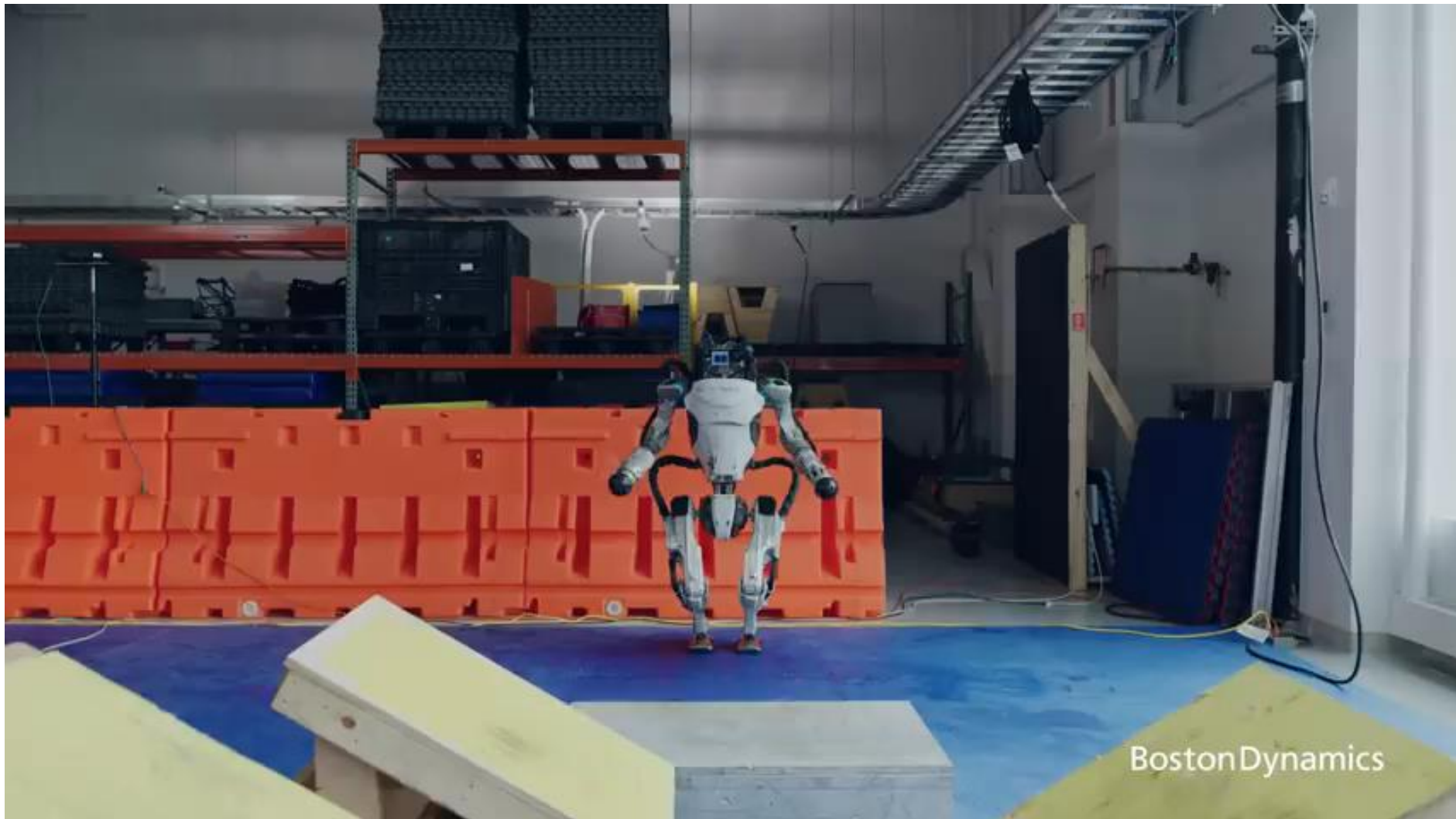


Machines that can learn from **data** and their **own experience**?

AI Around Us



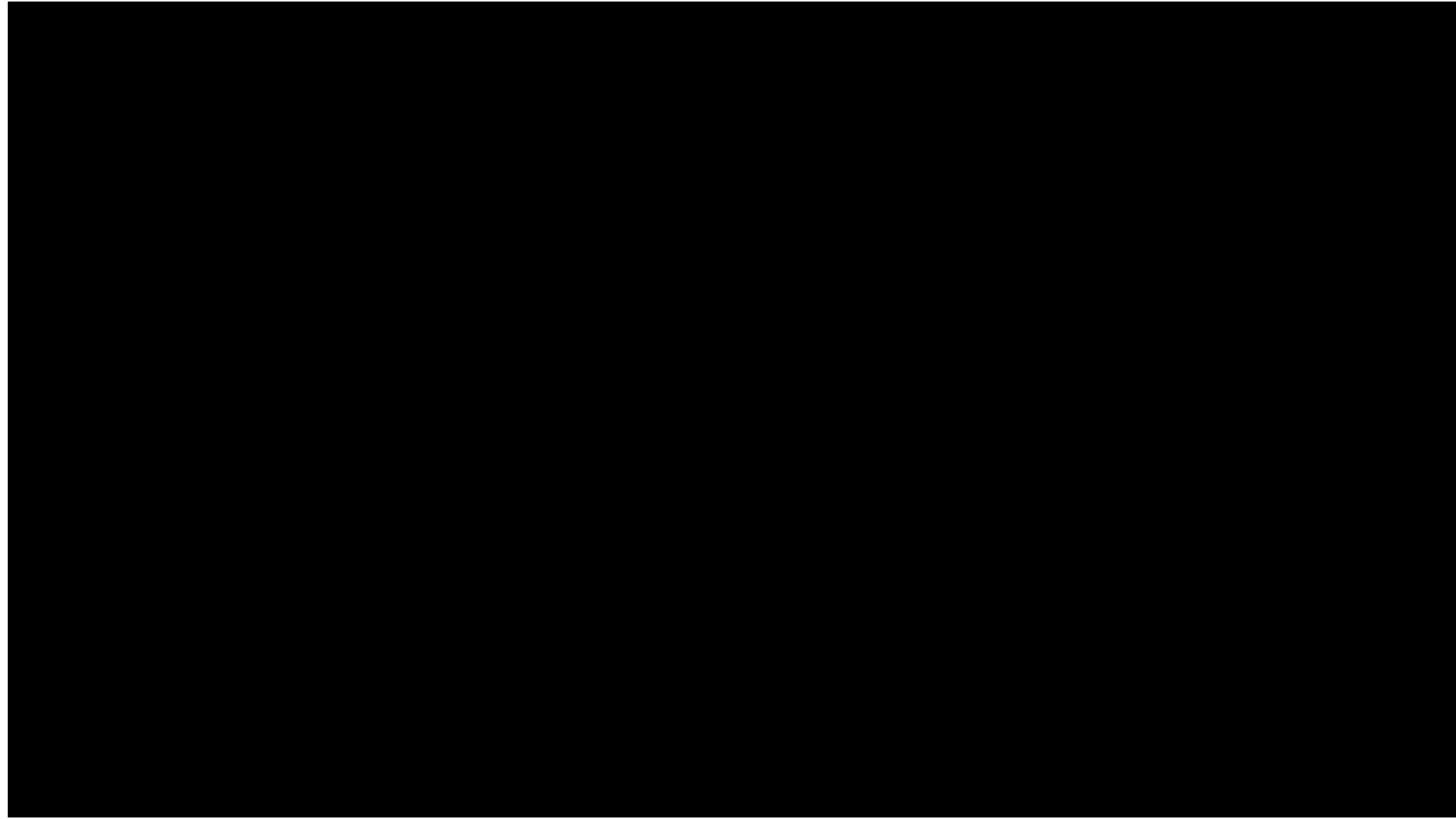
Boston Dynamics



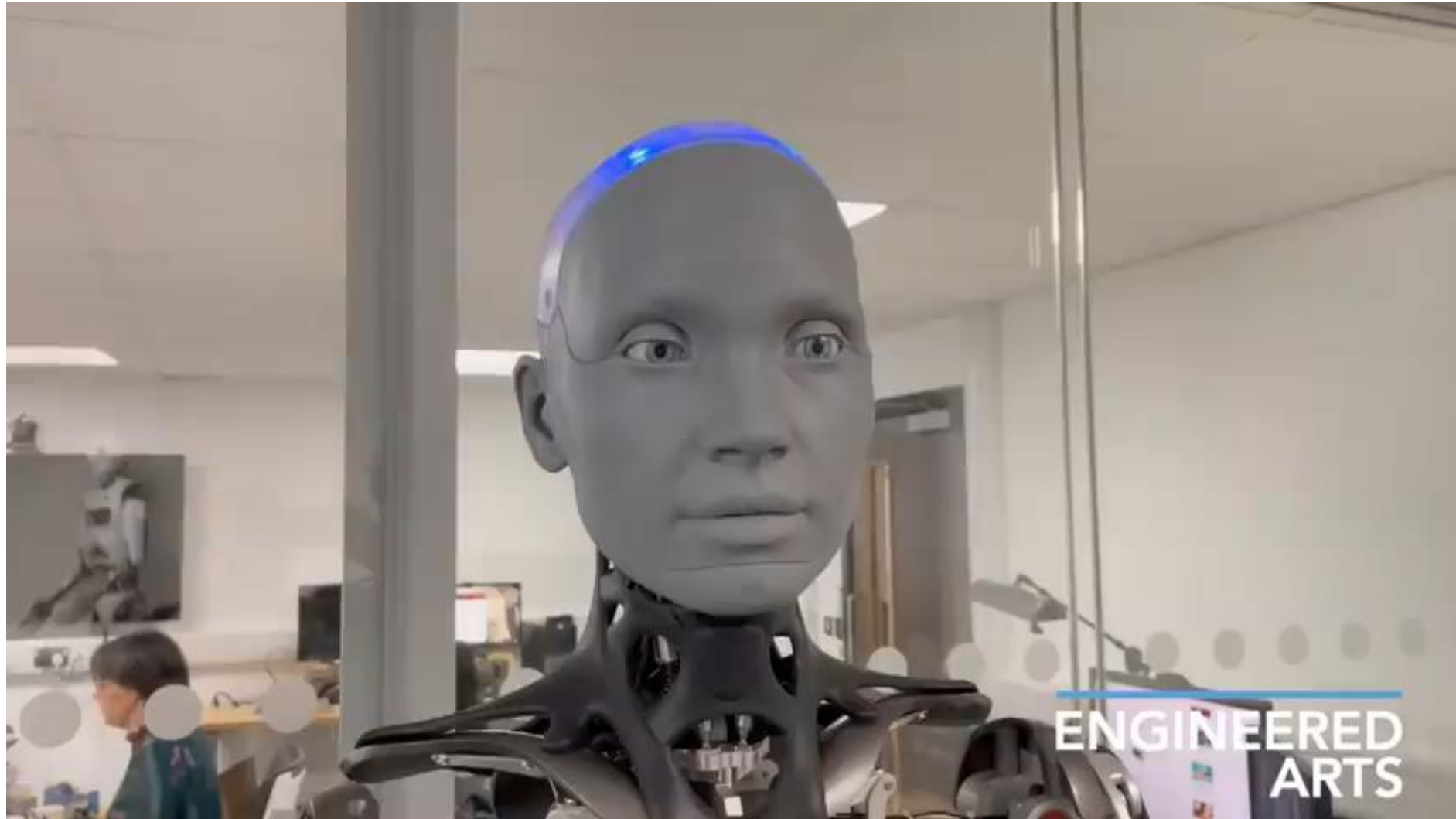
Evolution



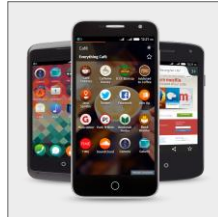
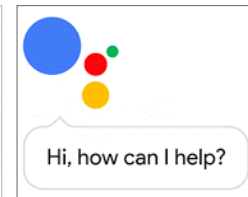
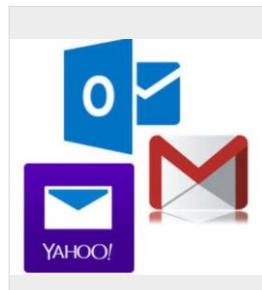
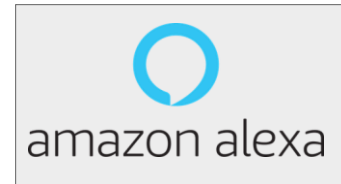
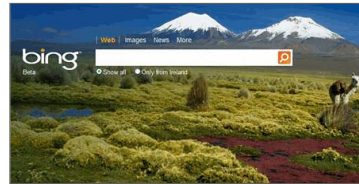
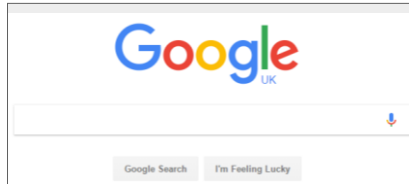
Sophia



Ameca



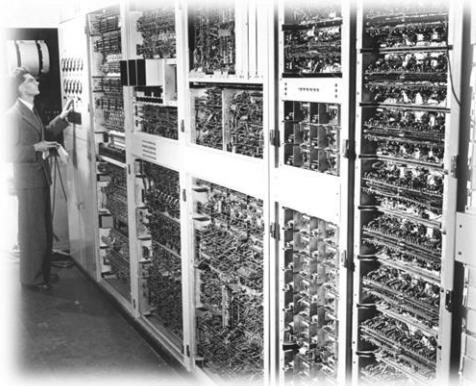
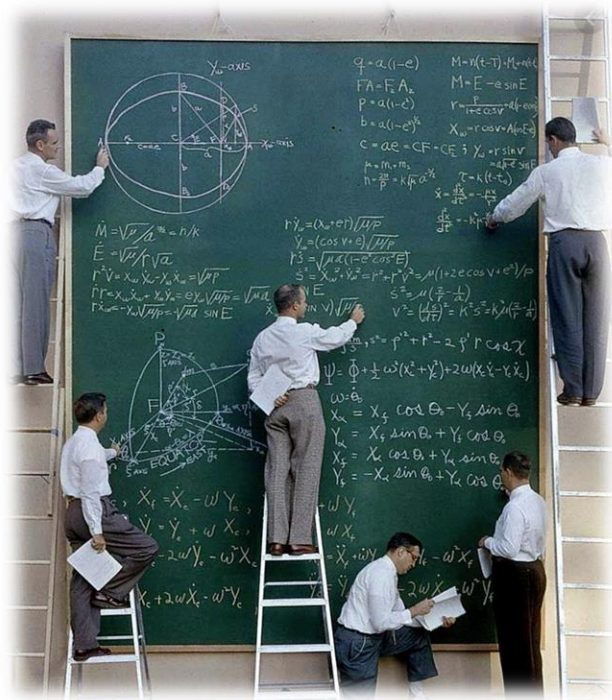
AI Around Us: No need to look so far away!



Machines as mechanical helpers



Machines as Intellectual helpers



But how is AI different from traditional Computer Science?



Traditional Problems

Play an audio/video file

Display a text file on screen

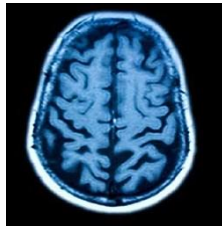
Perform a mathematical operation on numbers

Sort an array of numbers

Search for a number in a list of numbers



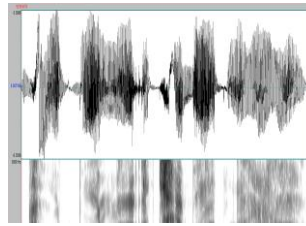
New Frontiers



Tumor? Y/N



Price?



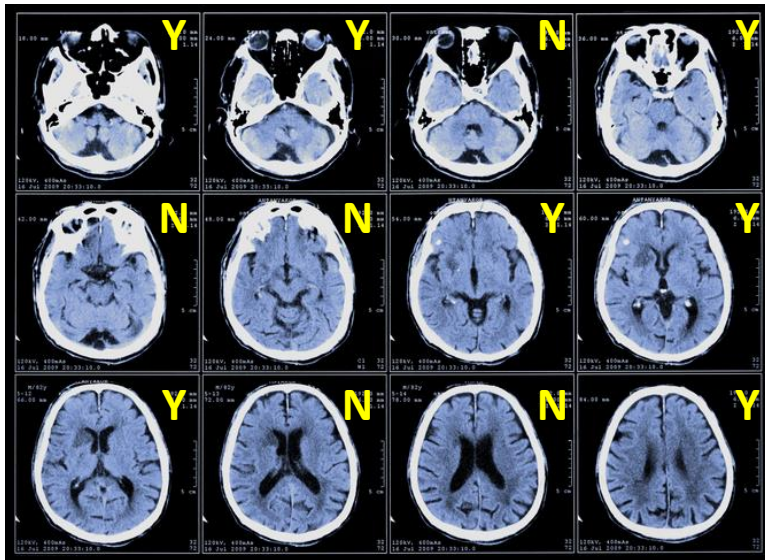
What was said?

A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell) A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell) A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell) A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchell)

Summarize text



Machine Learning



➤ \$150,000



➤ \$190,000



➤ \$350,000



➤ \$550,000



➤ \$90,000

Traditional CS



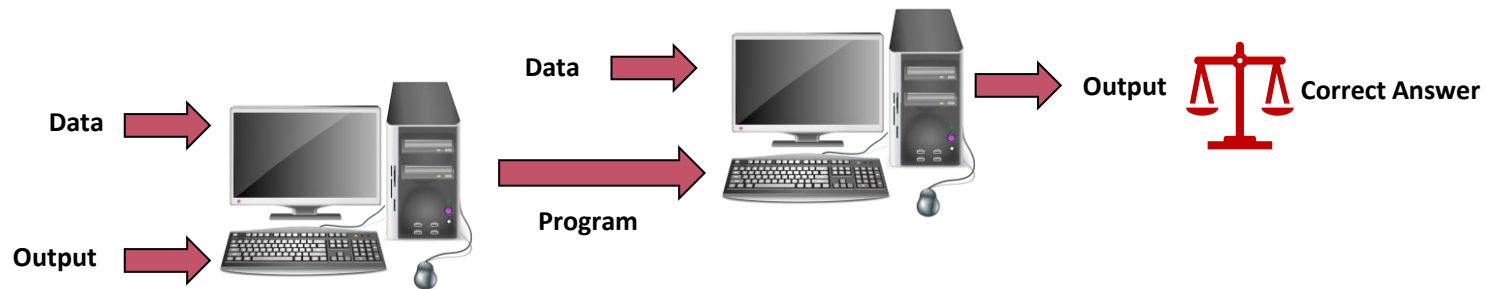
Machine Learning



Machine Learning Pipeline

Machine Learning

Traditional CS



Training

Testing

Evaluation

Machine Learning: Definition

A computer program **A** is said to learn from experience **E** with respect to some class of tasks **T** and performance measure **P** if its performance at tasks in **T**, as measured by **P**, improves with experience **E**. (Tom Mitchell, 1997)



Informally: *Algorithms that improve on some task with experience.*

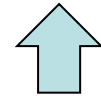
Moral: Don't get bogged down with tech lingo!

What's the difference between AI and ML?



Artificial Intelligence vs Machine Learning

- **The Historical Perspective:** The AI Winter
 - Unrealistic expectations
- **AI:** Make the machine more like a human
 - Give the machine a lot of world knowledge
 - A logical decision-making frame-work
- **ML:** Make a better machine – not necessarily emulating a human
 - Based on Statistics and Optimization
 - Learn from labelled data



AI



ML



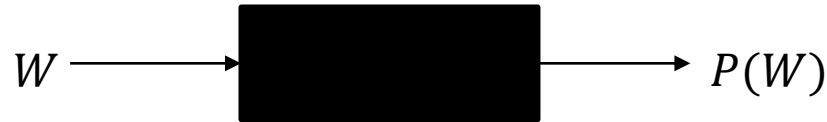
So, what has changed recently?



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What's the buzz all about?

A language model assigns probabilities to sentences
using data



Large Language Models

Chatbots and Conversational Agents

Reinforcement Learning from Human Feedback (RLHF)

Semantic Spaces – where words, images, and sounds
co-exist



At the heart of every intelligent machine
is a Classifier!



Download more graphics at www.psdgraphics.com

So, what's a Classifier?

- Well, a Classifier... Classifies!



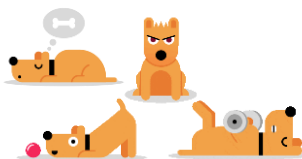
- A classifier classifies its inputs into a set of classes

- When the classes are predefined, we call it **“Supervised Machine Learning”**
 - In this case we need to tell the classifier what the classes are and provide some examples

CATS



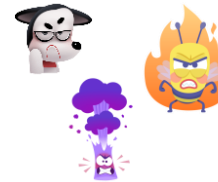
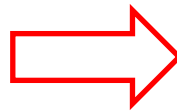
DOGS



?



- When the classes are not predefined, we call it **“Unsupervised Machine Learning”**
 - In this case the classifier just clusters the inputs into a bunch of classes without naming them.



Classifiers



Is this a cat or a dog?
{Cat, Dog}



CAT!



Should I hire this person?
{Yes, No}



NO!

Classifiers



{cat, dog}



cat



{happy, sad, angry,
surprised, neutral,
other}



happy

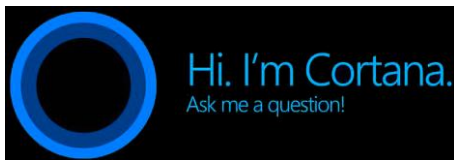
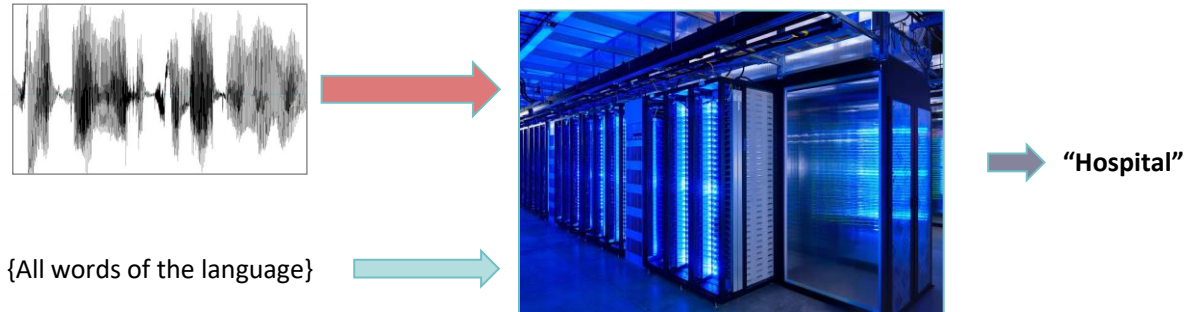


{empty, full}



empty

Classifiers



Classifiers



Return the *best* of {all board positions one black move from current}



0		
	x	



0		
x	x	

Return the *best* of {all positions one x move from current}

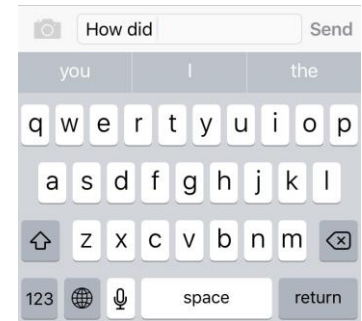


Classifiers: Language Models (LM)

I turned my homework ____



→ "in"



What is the most probable next word?
{Given all words of the language}



1. I turned my homework in.
2. I turned my homework fish.
3. ...

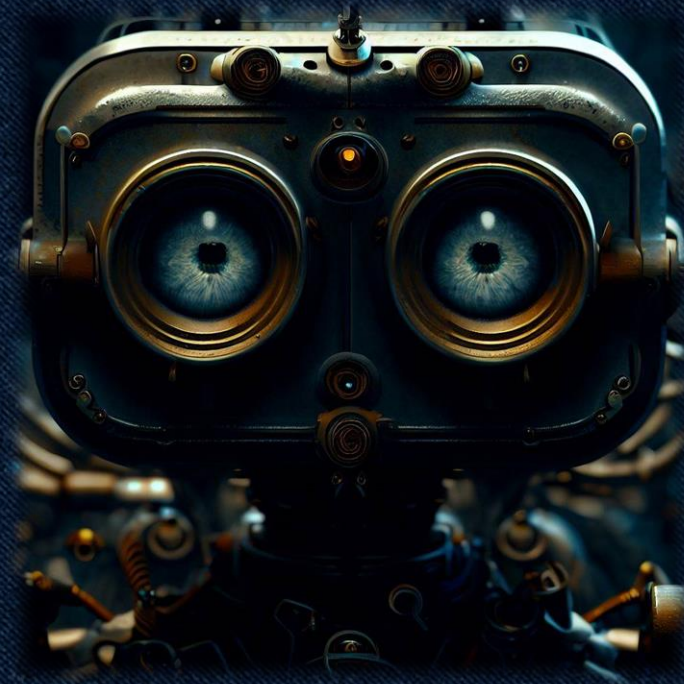


→ I turned my homework in.

Return the most probable sentence.



FEATURES: The eyes and ears of a classifier



Download more graphics at www.psdgraphics.com

What does a classifier see?

- Whatever you want it to see!
 - Features
- Donut vs. Bagel classifier



Feature
Extractor

Eyes: 2
Nose: 1
Nostrils: 2
Ears: 2

Fur Color: Brown and White
Eye Color: Green
Background Color: Black
Nose Color: Red

Image colors: Brown, Yellow,
Green, Black, Red,...

Color of pixel 0: Black
Color of Pixel 1: Black
....
Color of Pixel 2,000,000: Yellow

- Features?
 - Size – radius, diameter, circumference – cm, mm, inches?
Big/small?
 - Weight – units – heavy vs. light
 - Salty: Yes/No
 - Sweet: Yes/No
 - Squishy: Yes/No
 - Tasty: Yes/No
 - Healthy: Yes/No
 - Color: {all possible colors}
 - Cute: Yes/No
- Features that matter
 - Size
 - Weight

Now YOU are a Classifier!

A night vs. day classifier

- Features?
 - Brightness/Darkness
 - Sun
 - Moon
 - Stars
 - Sky color
 - Animals that come out during the day vs night

A Night vs. Day Classifier



A Night vs. Day Classifier



A Night vs. Day Classifier



How about Unsupervised Learning?

Unsupervised Learning

Goal: To find underlying patterns in the dataset. This process is known as **Clustering**.

What subpopulations exist in the following images?

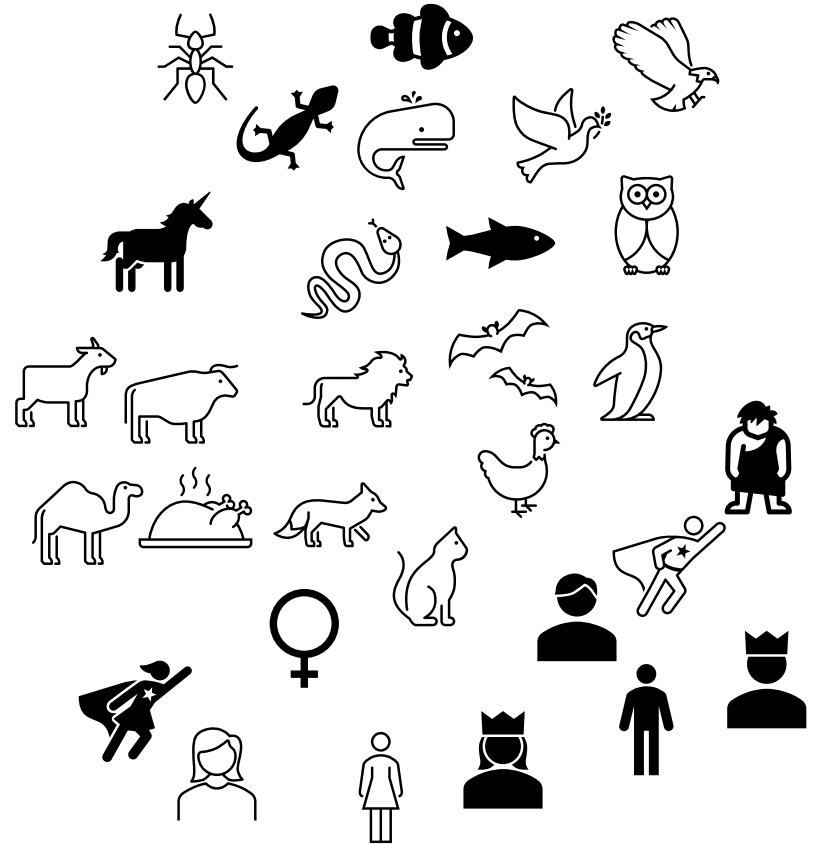


Important Questions:

- Are these subpopulations cohesive?
- Are there any outliers?
- Are they balanced?
- Is there a **hierarchy**?

Unsupervised Learning

- Find the underlying structure of the data or patterns in the data
 - Not about prediction
- What subpopulations exist in the data?
 - How many? And how big? How cohesive? Outliers?
 - Do elements in each subpopulation have commonalities?
 - Is there a hierarchy?



Data



Data – Big, Big... Data!

How do we obtain these massive datasets to train our Machine Learning models?

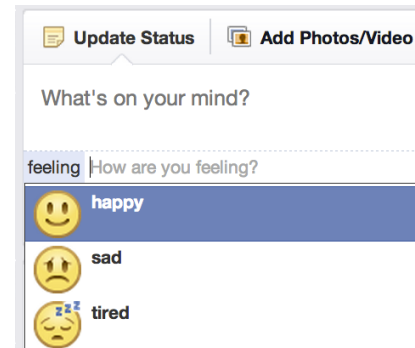
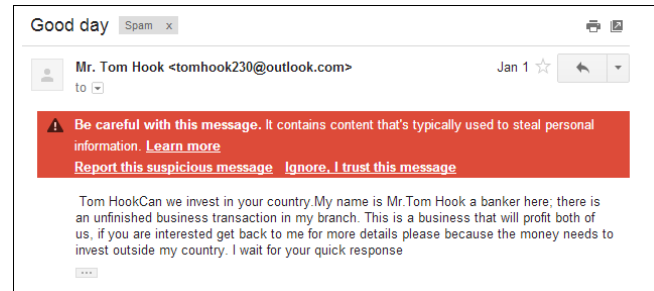
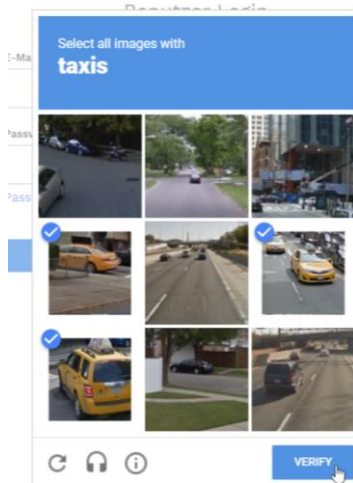
From real interactions e.g., call centers

Expert annotators e.g., hired teams of annotators

Crowd sourcing

Recaptcha:

Tagging:



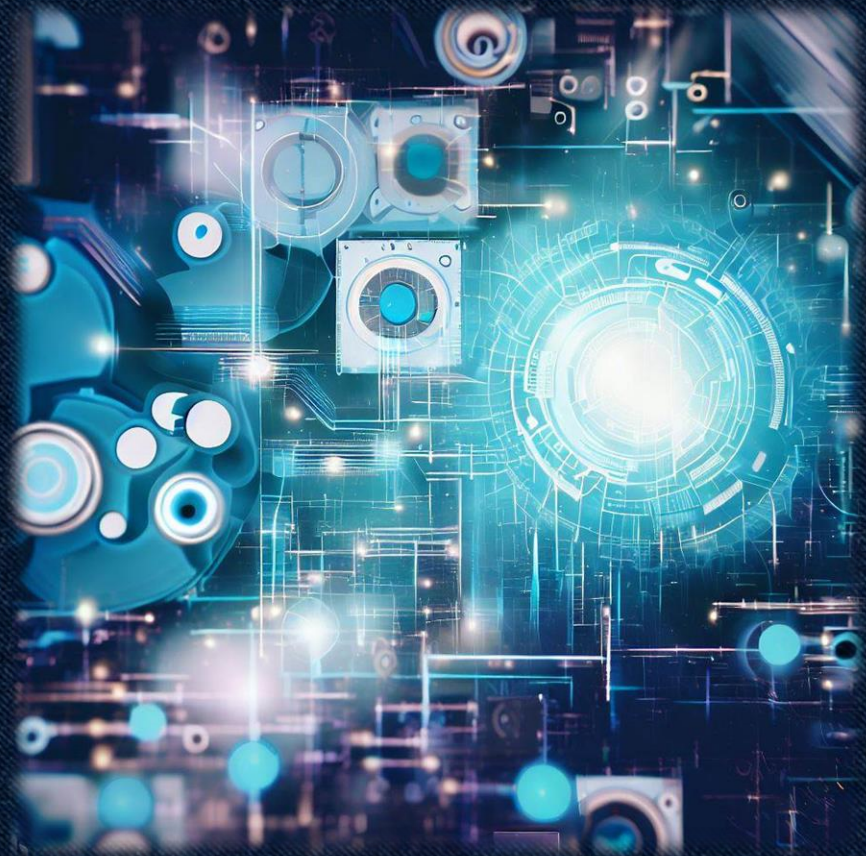
We tag data for “free” for using “free” services



SO MUCH OF "AI" IS JUST FIGURING OUT WAYS TO OFFLOAD WORK ONTO RANDOM STRANGERS.

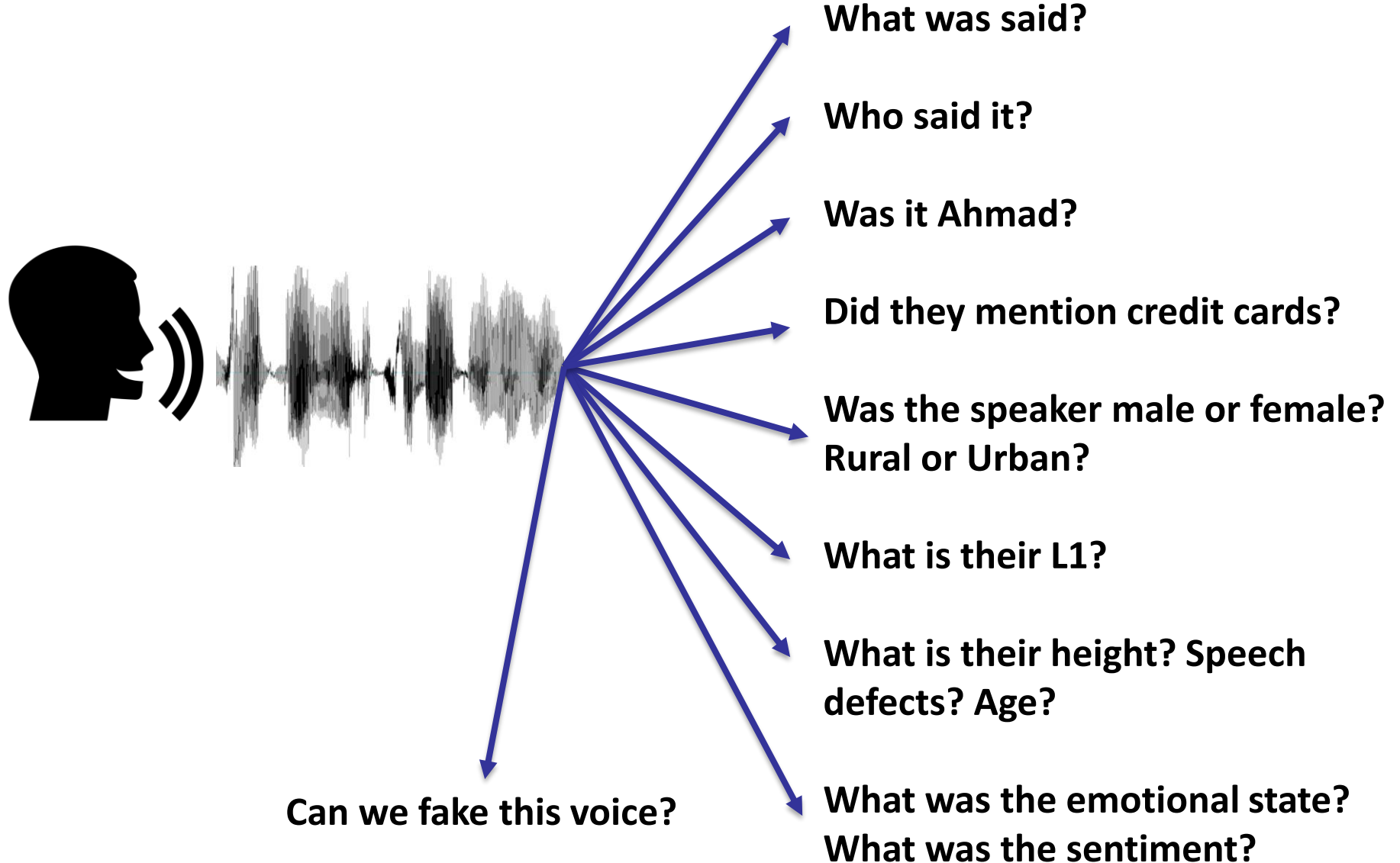
XKCD Data Science: <https://towardsdatascience.com/12-xkcd-strips-that-show-the-truth-about-ai-e09fbcd00c4c>

Applications

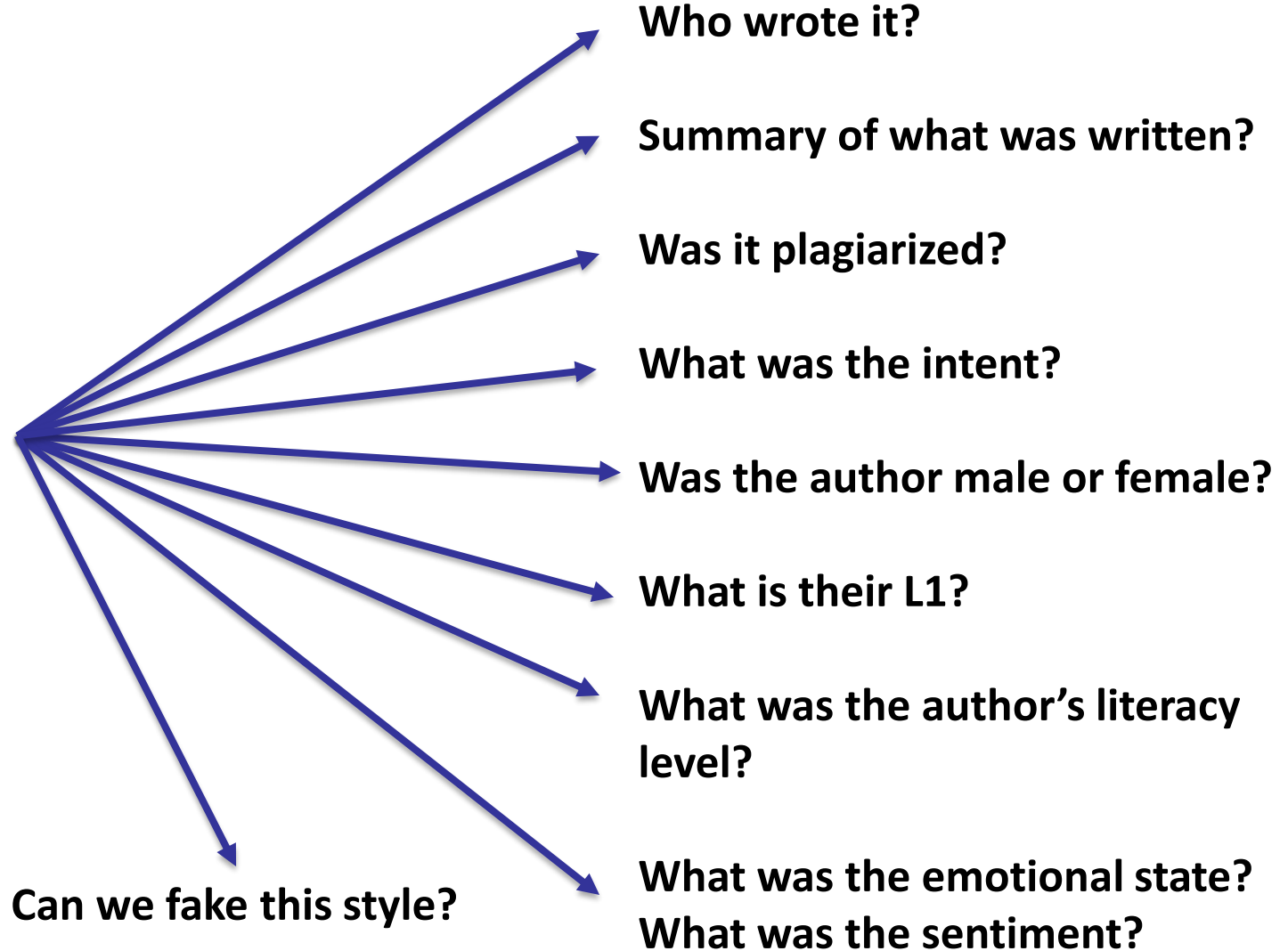


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Speech Technologies



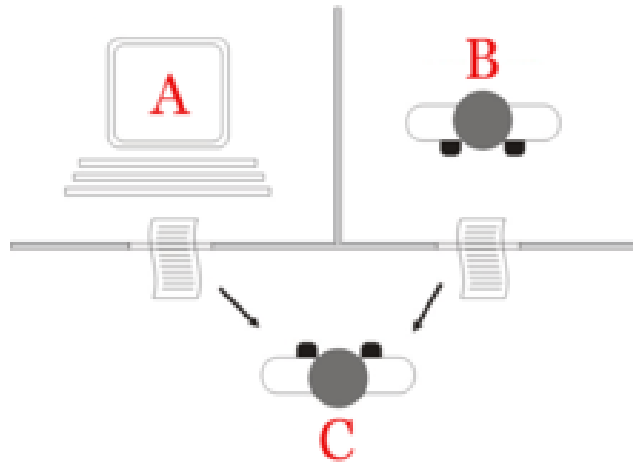
Text Technologies



Evaluation: Ideal and Practical

The Turing Test

- The "standard interpretation" of the Turing test:
 - Player C, the interrogator, tries to determine which player – A or B – is a computer and which is a human.
 - The interrogator is limited to using the responses to written questions to make the determination.



- https://en.wikipedia.org/wiki/Turing_test
- XKCD Data Science: <https://towardsdatascience.com/12-xkcd-strips-that-show-the-truth-about-ai-e09fbcd00c4c>

ML in Local Languages of Pakistan

Speech Recognition (Speech to Text)

3s

[106] import IPython
audio_file = "test.wav"
audio, sr = get_audio()
IPython.display.Audio(audio_file)

Saving the recording... pls wait!

▶

0:00

🔊

0s

[107] !ffmpeg -y -i test.wav -ar 16000 -ac 1 demo.wav

ffmpeg version 3.4.8-0ubuntu0.2 Copyright (c) 2000-2020 the FFmpeg developers
built with gcc 7 (Ubuntu 7.5.0-3ubuntu1~18.04)
configuration: --prefix=/usr --extra-version=0ubuntu0.2 --toolchain=hardened --libdir=/usr/lib/x86_64-linux-gnu --incdir=/usr/include/x86_64-linux-gnu --enable-gpl --disable-stripping --enable-avresample --ena
libavutil 55. 78.100 / 55. 78.100
libavcodec 57.107.100 / 57.107.100
libavformat 57. 83.100 / 57. 83.100
libavdevice 57. 10.100 / 57. 10.100
libavfilter 6.107.100 / 6.107.100
libavresample 3. 7. 0 / 3. 7. 0
libswscale 4. 8.100 / 4. 8.100
libswresample 2. 9.100 / 2. 9.100
libpostproc 54. 7.100 / 54. 7.100
Guessed Channel Layout for Input Stream #0.0 : mono
Input #0, wav, from 'test.wav':
Duration: 00:00:00.30, bitrate: 769 kb/s
Stream #0:0: Audio: pcm_s16le ([1][0][0][0] / 0x0001), 48000 Hz, mono, s16, 768 kb/s
Stream mapping:
Stream #0:0 -> #0:0 (pcm_s16le (native) -> pcm_s16le (native))
Press [q] to stop, [?] for help
Output #0, wav, to 'demo.wav':
Metadata:
ISFT : Lavf57.83.100
Stream #0:0: Audio: pcm_s16le ([1][0][0][0] / 0x0001), 16000 Hz, mono, s16, 256 kb/s
Metadata:
encoder : Lavc57.107.100 pcm_s16le
size= 9kB time=00:00:00.30 bitrate= 258.1kbits/s speed= 397x
video:0kB audio:9kB subtitle:0kB other streams:0kB global headers:0kB muxing overhead: 0.812500%

0s

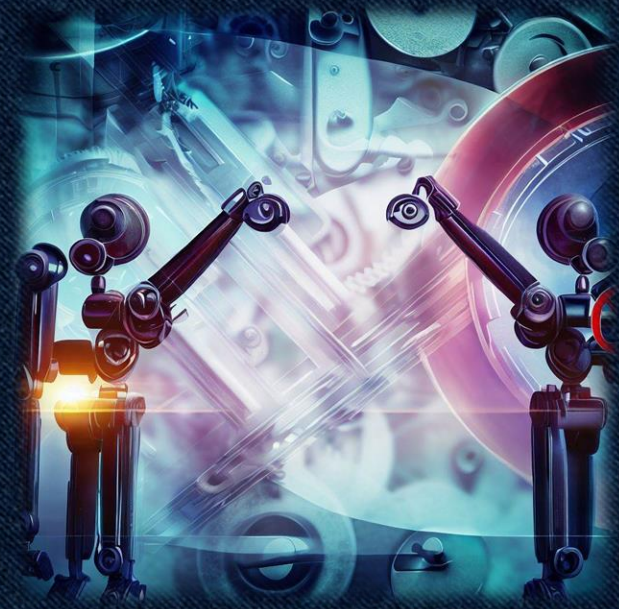
transcribe("demo.wav")

Transcription ---



Speech Synthesis (Text to Speech)

Limitations and Challenges



Download more graphics at www.psdgraphics.com

Challenges of ML – Explainability

- A classifier can potentially learn to classify on the basis of features not desirable for humans
 - All dogs wearing a collar in the training data while no cat is wearing it – ML just learns to separate based on collar
 - All horse images have a copyrights notice – ML just learns to recognize horses based on the copyrights notice

Explainable ML: The results should be understandable by humans

- As opposed to a black-box system

Challenges – Fairness in AI

- AI tends to reflect the biases of the society
 - Human taggers who mark a recording as misinformation based on accent or gender
 - Court decisions in country that make a rich person's acquittal more likely
 - Automated standardized testing in the US could yield unfavorable results for certain demographic groups
 - AI plays a deciding role in hiring decisions, with up to 72% of resumes in the US never being viewed by a human
 - Decisions on immigration, bank loans, credit history checks, criminal profiling

Machine Learning in Low-resource settings

- Problems where large data sets and tools are not available
- Natural Language Processing and Speech problems for languages of developing regions
 - Pakistan has 71 languages
 - We barely have speech recognition capabilities for Urdu

Why is this important?

The Internet

- The internet has transformed the way people participate in the information ecology and digital economy
 - Social media, online discussion forums, crowdsourcing marketplaces
- The Internet empowers people who enjoy access to it
 - Mostly urban, affluent and literate

So, who is left out?

Oral and Offline

- **2.9 billion people worldwide are offline**
 - That is **37%** of the world population
 - Of these, **96%** live in developing countries.
 - **10%** of the developed world, **43%** of the developing world and **73%** of the Least Developed Countries are offline*
- **Offline populations**
 - too poor to afford Internet-enabled devices
 - too remote to access the Internet
 - too low-literate to navigate the mostly-text-driven Internet

References: International Telecommunication Union (ITU): Facts and Figures 2021: 2.9 billion people still offline, [Link](https://www.itu.int/en/ITU-D/Statistics/Documents/facts/FactsFigures2021.pdf), <https://www.itu.int/en/ITU-D/Statistics/Documents/facts/FactsFigures2021.pdf>, last accessed Feb 22, 2022
McKinsey (2014), WHO, World Bank, Ethnologue, The World Fact book – CIA, GSMA Mobile Economy, weforum.org

Digital Divides

- **Gender Divide:** More men than women use the Internet.
 - The gap is smaller in developed countries and larger in developing countries, and LDCs (4 out of every 5 women are offline in LDCs).
- **Urban-Rural Divide:** More urban than rural people use the internet
 - Globally, people in urban areas are twice as likely to use the Internet than those in rural areas (47% vs 13% in LDCs).
- **Literacy Divide:** The (mostly) text-driven internet is not suitable for:
 - The 2.2 billion visually impaired individuals
 - Low-literates, oral cultures, native speakers of unwritten languages (47% of all languages)

References: WHO ([link](#)), McKinsey (2014), WHO, World Bank, Ethnologue, The World Fact book – CIA, GSMA Mobile Economy, ITU, <https://itu.foleon.com/itu/measuring-digital-development/gender-gap/>, PTA, <https://www.pta.gov.pk/en/telecom-indicators>

Lack of access to Information and Connectivity can be a major impediment to Development



Managing Expectations

- **Too optimistic/eager**

- AI replacing humans as caring partners, AI replacing creative professions, super-intelligent AI
- AI invasion, robots gaining sentience and self-awareness, technological singularity, terminators!



Managing Expectations

- **Too pessimistic/reluctant**

- No practical applications of AI, no chances of a positive social impact, AI cannot help with anything
- No expected social harm of the mistakes of AI, no associated risks of integrating AI irresponsibly



- The differences between these extremes have been responsible for the AI winters



TECHNOLOGY

Google shares drop \$100 billion after its new AI chatbot makes a mistake

February 9, 2023 · 10:15 AM ET

By [Emily Olson](#)

<https://www.npr.org/2023/02/09/1155650909/google-chatbot-error-bard-shares>

Managing Expectations

- **Just right**

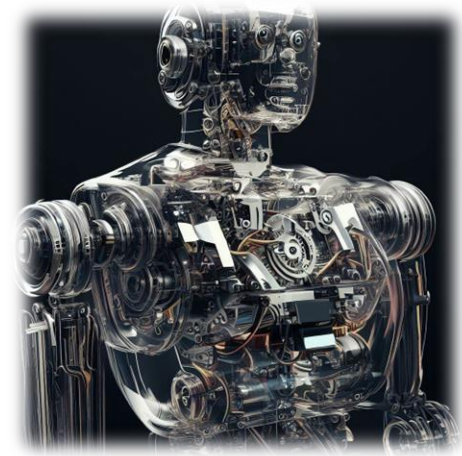
- AI can aid human efforts and professions, it can transform the industry, improve performance, precision and accuracy, it can produce meaningful impact on the society
- Real harms that must be mitigated e.g., disruptive influence on the job market, hard to explain and interpret, biases, the need for fairness and regulation, fail-safes to mitigate potential harms



Responsible AI

Artificial Intelligence should be practiced in a manner that is:

- Explainable/Transparent/Interpretable
- Fair/unbiased
- Ethical
 - Disruptive influences
 - Privacy and informed consent
- Safe
 - For the stakeholders
 - From adversarial attacks (the cat-n-mouse game)
- Regulated





IN CS, IT CAN BE HARD TO EXPLAIN
THE DIFFERENCE BETWEEN THE EASY
AND THE VIRTUALLY IMPOSSIBLE.

- XKCD Data Science: <https://towardsdatascience.com/12-xkcd-strips-that-show-the-truth-about-ai-e09fbcd00c4c>

For more details please visit
<http://aghaaliraza.com>



Thank you!